GDC Resolve Encoder

User guide

Free H.264 and H.265 export plugin for DaVinci Resolve Studio on Mac

Version 1.8.0 · https://gordas.dev/gdc-resolve-encoder/

GDC Resolve Encoder · Dumitru Cristinel Gordas (GDC)

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1. What it is

GDC Resolve Encoder adds complete H.264 and H.265 encoders (x264, x265 and Apple VideoToolbox), three containers of its own (GDC QuickTime, GDC MP4, GDC Matroska) and HDR10 metadata in the file to DaVinci Resolve Studio, right inside the Deliver page.

Edition 1.8.0 for macOS is **free and open source**: no account, activation, trial period or donation is required. It works only with DaVinci Resolve Studio; the free Resolve does not load export plugins. The plugin runs locally and sends no data anywhere.

2. Installation

Requirements: macOS 26 or later (declared minimum; verified on macOS 27), a Mac with Apple Silicon, DaVinci Resolve Studio.

- Download **GDCResolveEncoder-1.8.0.pkg** from the project page and double-click it.
- Follow the installer (accept the license; it asks for your Mac password).
- Restart DaVinci Resolve. Nothing else needs installing: FFmpeg and the encoding libraries ship inside the package.

The package installs to:

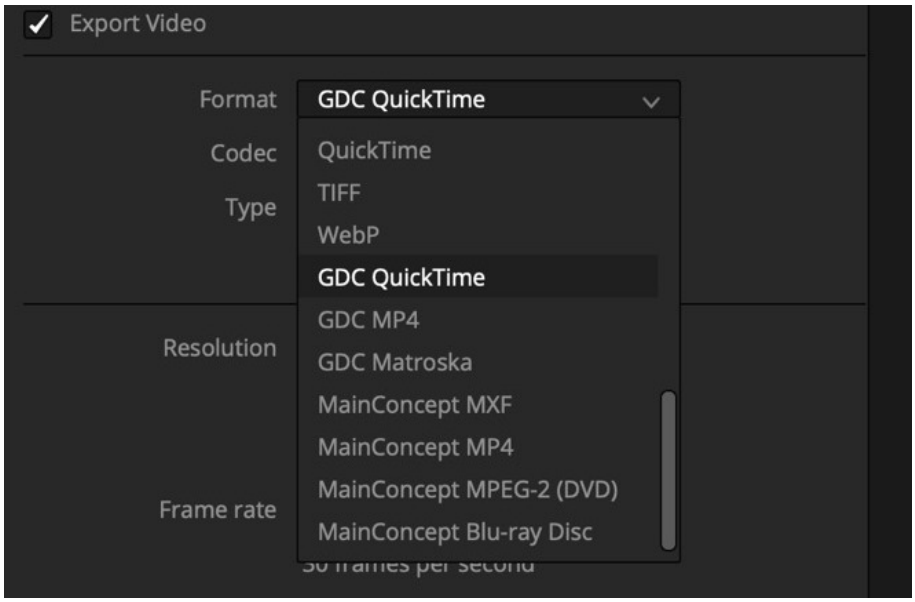
```
/Library/Application Support/Blackmagic Design/DaVinci
Resolve/IOPlugins/gdc_resolve_encoder.dvcp.bundle
```

To uninstall, quit Resolve and move the folder above to the Trash (Finder asks for your password).

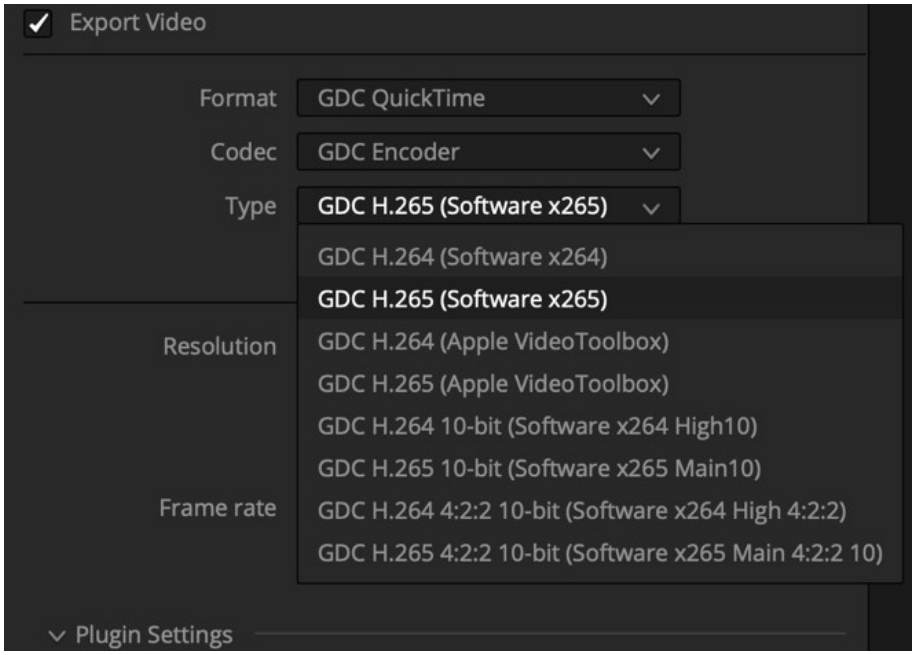
3. The Deliver flow

In the Deliver page, Render Settings, Video tab:

- **Format**: pick GDC QuickTime, GDC MP4 or GDC Matroska (they appear after the native formats).
- **Codec**: pick GDC Encoder.
- **Type**: pick one of the 8 variants (table below).
- **Plugin Settings**: tune the quality, then add the job to the render queue as usual.



The Format list: GDC QuickTime, GDC MP4 and GDC Matroska.



Codec = GDC Encoder; the Type list with the 8 variants.

4. The Type variants

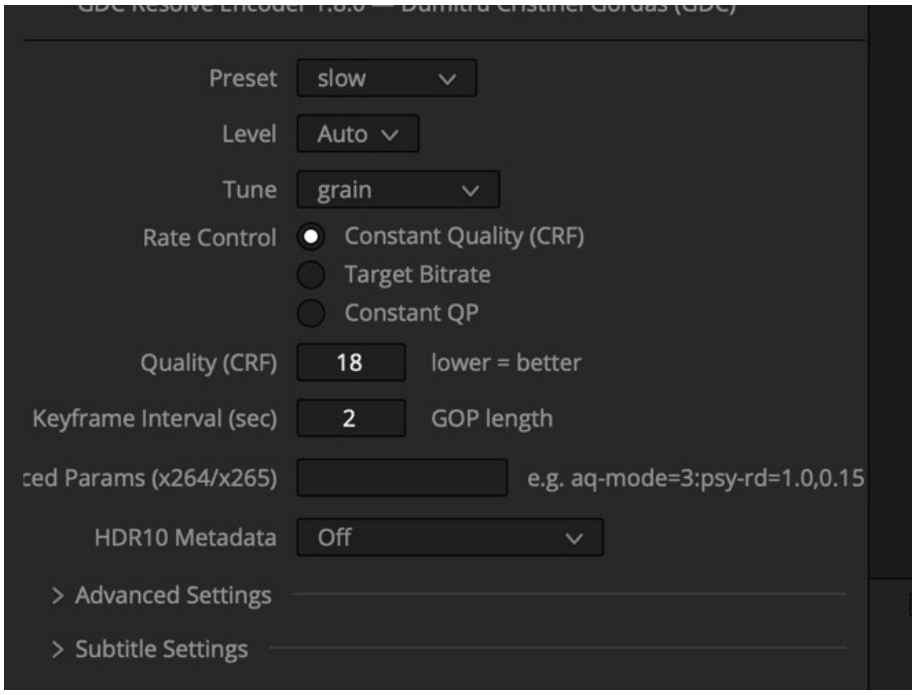
Type variant	Encoder	Depth / chroma
GDC H.264 (Software x264)	libx264	8-bit 4:2:0
GDC H.265 (Software x265)	libx265	8-bit 4:2:0
GDC H.264 (Apple VideoToolbox)	hardware	8-bit 4:2:0
GDC H.265 (Apple VideoToolbox)	hardware	8-bit 4:2:0
GDC H.264 10-bit (x264 High10)	libx264	10-bit 4:2:0
GDC H.265 10-bit (x265 Main10)	libx265	10-bit 4:2:0
GDC H.264 4:2:2 10-bit (High 4:2:2)	libx264	10-bit 4:2:2
GDC H.265 4:2:2 10-bit (Main 4:2:2 10)	libx265	10-bit 4:2:2

Hardware variants appear in the list only if your Mac can actually run them. HDR10 Metadata is available on the software variants.

5. The panel settings

Setting	What it does
Preset	Speed / efficiency trade-off, from ultrafast to veryslow. Slower = smaller file at the same quality.
Level	H.264/H.265 level (3.0 – 5.2) or Auto. Leave Auto unless an old player requires otherwise.
Tune	Content-specific tuning: film, animation, grain, stillimage, psnr, ssim, fastdecode, zerolatency (the list differs slightly between H.264 and H.265).
Rate Control	Constant Quality (CRF), Target Bitrate or Constant QP. CRF is the usual choice: constant quality, variable size.
Quality (CRF)	0 – 51, lower = better. Starting point: 18 for masters, 20–23 for web delivery.
Quality (1–100)	On VideoToolbox, in quality mode: higher = better. Starting point: 65.

Setting	What it does
Keyframe Interval	Distance between keyframes, in seconds (default 2).
Advanced Params	Expert x264/x265 parameters, for example aq-mode=3:psy-rd=1.0,0.15. They take precedence over the settings above.
HDR10 Metadata	Off or On (PQ exports only): mastering primaries (P3-D65 or Rec.2020), peak luminance 100–10000 nits, MaxCLL, MaxFALL (0 = not signalled).



The Plugin Settings panel for GDC H.265 (Software x265).

6. Containers and metadata

- The three containers write the color tags (primaries, transfer, matrix, range) into the file.
- On PQ exports with HDR10 on, the file also carries mastering display and MaxCLL/MaxFALL, both in the video stream and in the container.
- Audio is written as uncompressed PCM (16, 24 or 32 bit), unchanged. In MP4, PCM audio has limited support in some players.
- Start timecode and markers are not written yet.
- Width and height must be even (for 4:2:2 only the width matters); otherwise the export is refused with a message in the log.

7. Starting recipes

Setting	What it does
10-bit master	GDC H.265 10-bit (Main10), GDC QuickTime, CRF 16–18, preset slow.
Web delivery	GDC H.264 (Software x264), GDC MP4, CRF 20–23, preset medium.
Quick preview	GDC H.265 (Apple VideoToolbox), Quality 65.
HDR10 (PQ)	GDC H.265 10-bit, GDC MP4 or QuickTime, HDR10 Metadata = On, primaries and luminance matching the mastering display.

These are starting points, not guarantees: check the result on your own material.

8. What was verified

Edition 1.8.0 was verified in DaVinci Resolve Studio 21.2 on an Apple Silicon Mac, on cinema camera material, with 17 exports (full clips and a multi-clip sequence, with sound):

- picture: minimum PSNR 40.6 dB against the source, frames counted correctly;
- sound: in sync (offset 0), samples identical to the source except for the few milliseconds of fade at clip edges;
- color: tags written correctly, HDR10 present in the file;
- reimport in Resolve and playback in FFmpeg without errors.

Windows is not part of this edition. The free Windows edition, with no activation, is v1.7.1 (maintenance); its current status is on the product page.

9. Common problems

- **The GDC formats do not appear:** make sure you use DaVinci Resolve Studio and restarted Resolve after installing; check that the install folder above exists.
- **A hardware variant is missing:** your Mac cannot run it; use the software variant.
- **Failed export:** in Resolve, Help → Create Diagnostics Log on Desktop; the log holds the reason and the rate-control settings used.
- **Cancelling:** a stopped export leaves a valid file, truncated at the last written frame.

10. License and credits

The plugin code is MIT-licensed. The Mac package includes FFmpeg, x264 and x265 and is distributed under the GPLv3; the license texts and the source offer are inside the package, in Contents/Resources/Licenses. Source code: https://github.com/gordasgdc/gdc-resolve-encoder.

Author: Dumitru Cristinel Gordas (GDC). Product page: https://gordas.dev/gdc-resolve-encoder/. GDC Resolve Encoder is an independent project; it is not developed, endorsed or supported by Blackmagic Design. DaVinci Resolve is a trademark of its owner. Contributions (especially for Windows) are welcome through Issues and Pull Requests.